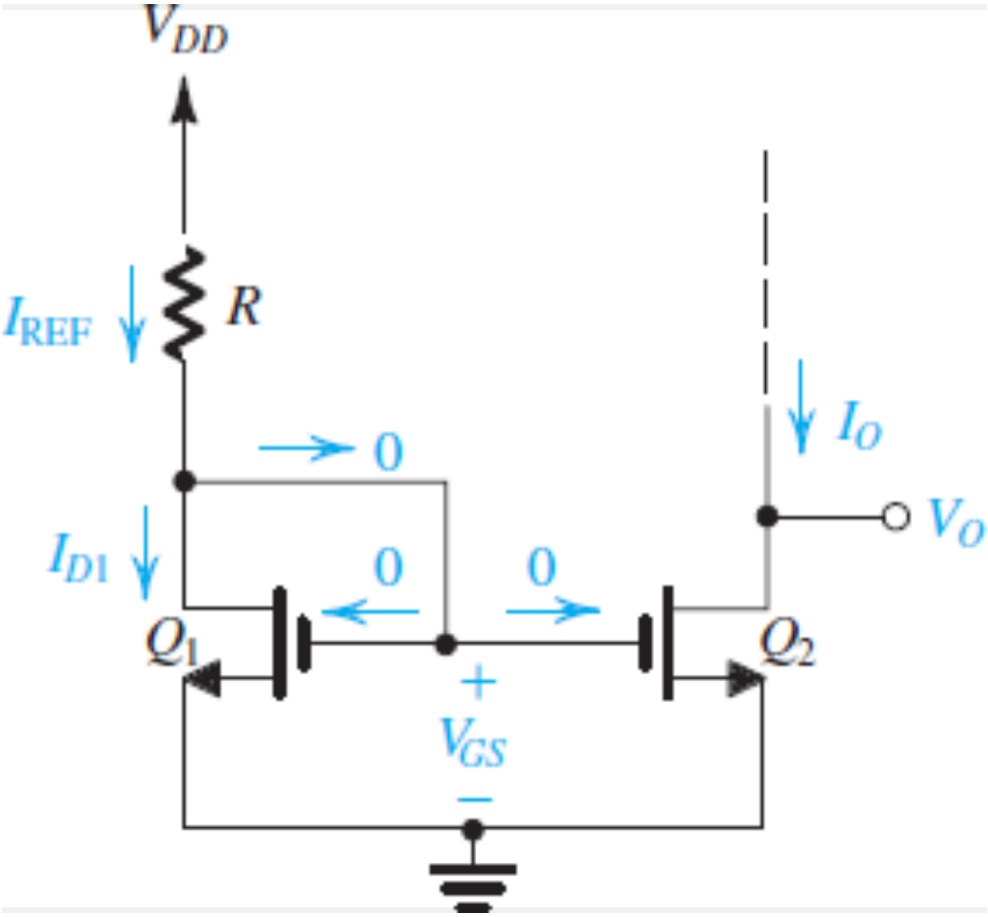
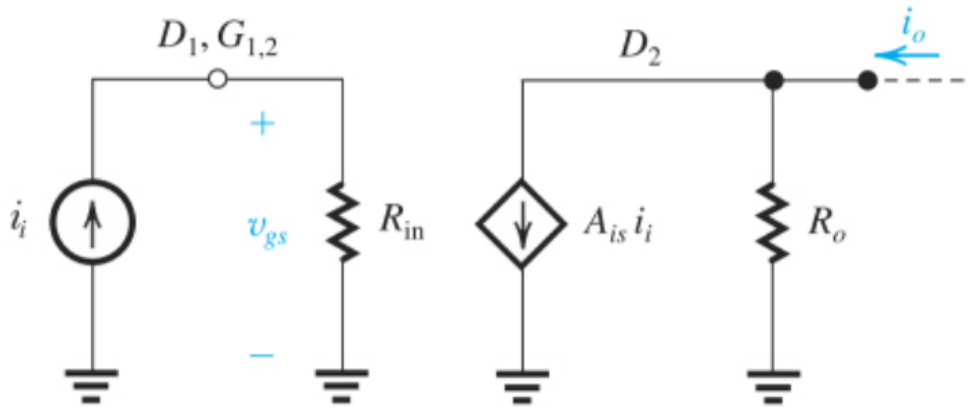
# MOSFET Current Source

## Schematic



### Small-Signal Model




(b)

# Equations

| Equation | Conditions | Notes |
|---|---|---|
| $I_{D1} = I_{REF} = \frac{V_{DD} - V_{GS}}{R}$ | | Gate currents are zero |
| $I_O = I_{D2} = \frac{1}{2} k'_n \left(\frac{W}{L}\right)_2 (V_{GS} - V_t)^2$ | $V_O \geq V_{GS} - V_t$ | Must operate in satur |
| $I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF} \left[1 + \frac{V_O - V_{GS}}{V_{A2}}\right]$ | | |
| $\Delta I_O = \frac{\Delta V_O}{r_{o2}} = \frac{V_O - V_{GS}}{r_{o2}}$ | | |
| $\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$ | Negligible channel-length modulation | |
| $I_O = I_{REF}$ | Identical transistors | |
| $A_{is} = \frac{g_{m2}}{g_{m1}} \frac{1}{1 + (1/g_{m1} r_{o1})}$ | | |
| $A_{is} = \frac{(W/L)_2}{(W/L)_1} \left[\frac{1}{1 + (1/g_{m1} r_{o1})}\right]$ | | |

# Notes

| Term | Description |
|---|---|
| $A_{is}$ | Small signal short-circuit current gain |